IE6600 Computation and Visualization for Analytics
Project 1 Report

U.S. Chronic Disease Indicators

Group 7
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1. Introduction

Healthcare is one of the most data-rich and impactful domains for analytical exploration. Chronic diseases — such as diabetes, cardiovascular disease, obesity, and asthma — are among the leading causes of death and disability in the United States, and understanding their patterns across states and over time is critical for informing public health policy.

For this project, we selected the U.S. Chronic Disease Indicators (CDI) dataset, published by the Centers for Disease Control and Prevention (CDC) and made available through data.gov. This dataset provides a standardized set of indicators that allow states and territories to uniformly define, collect, and report chronic disease data. It covers a wide range of topics including cancer, nutrition, alcohol use, mental health, and more — spanning multiple years and all U.S. states and territories.

Our motivation for choosing this dataset was a shared interest in exploring the healthcare sector through a data-driven lens. We wanted to understand how chronic disease rates vary across the country, whether they have changed meaningfully over time, and which disease categories carry the greatest burden on the population.

The primary objectives of this project are:

- To clean and prepare a large, real-world public health dataset for analysis.
- To use **matplotlib** to build static visualizations that reveal geographic, temporal, and categorical patterns in chronic disease data.
- To apply probability and statistical methods — including hypothesis testing, distribution analysis, and regression — to draw meaningful, evidence-backed conclusions.

The remainder of this report documents our full analytical process: from data acquisition and cleaning, through exploratory data analysis, to statistical analysis and advanced modeling.

2. Data Acquisition and Inspection

2.1 Data Acquisition

The dataset was obtained from the U.S. Centers for Disease Control and Prevention (CDC) via data.gov. It is titled U.S. Chronic Disease Indicators (CDI) and is publicly available for download in CSV format. The dataset was loaded directly into our Python analysis environment using the pandas library:

```
df = pd.read_csv("U.S._Chronic_Disease_Indicators.csv")
```

2.2 Dataset Structure

Upon loading, the dataset was found to contain 309,215 rows and 34 columns, representing chronic disease indicator records across all U.S. states and territories from 2015 to 2022. The key columns in the dataset are described below:

Column	Description
YearStart / YearEnd	The year the indicator data was collected
LocationDesc / LocationAbbr	Full U.S. state or territory name and abbreviation
Topic	Disease category (e.g., Diabetes, Cancer, Obesity)
Question	The specific indicator being measured
DataValue	The reported numeric value for the indicator
LowConfidenceLimit / HighConfidenceLimit	95% confidence interval bounds
DataSource	The source of the data (e.g., BRFSS, YRBSS)
Stratification1	Population subgroup (e.g., Overall, Male, Female)

2.3 Data Types

The dataset contains a mix of data types. Numeric columns include YearStart, DataValue, LowConfidenceLimit, and HighConfidenceLimit. The remaining columns are object (string) types representing categorical fields such as LocationDesc, Topic, Question, and DataSource. Some columns, such as Geolocation, were stored as strings despite representing spatial data — a type inconsistency that was addressed during cleaning.

2.4 Statistical Summary

A statistical summary of the cleaned dataset using df.describe() revealed that the DataValue column has a mean of 41.18 and a median of 23.95, indicating a moderately right-skewed distribution. The maximum value of 1,456.5 reflects high-burden indicators in specific subpopulations, such as hospitalisation rates reported per 100,000. The confidence limit columns (LowConfidenceLimit mean: 36.83, HighConfidenceLimit mean: 46.03) are closely aligned with the DataValue mean, confirming the dataset's internal consistency after cleaning.

Statistic	YearStart	DataValue	LowConfidenceLimit	HighConfidenceLimit
Count	188,885	188,885	188,885	188,885
Mean	2020.08	41.18	36.83	46.03
Std Dev	1.50	67.27	65.03	70.00
Min	2015	0.00	0.00	0.00
25%	2019	11.50	9.10	14.00
50%	2020	23.95	19.50	29.20
75%	2021	46.90	40.30	54.30
Max	2022	1,456.50	1,427.00	1,485.90

The cleaned dataset spans from **2015 to 2022** and covers all U.S. states and territories. The mean DataValue of **41.18** is much closer to the median of **23.95** after removing extreme outliers during cleaning, indicating a moderately right-skewed distribution with a max of 1,456.5 reflecting high-burden indicators in specific subpopulations.

3. Data Cleaning and Preparation

Meticulous data cleaning was executed to ensure the dataset's integrity and reliability for subsequent statistical and visual analysis. The following steps were taken systematically.

3.1 Handling Missing Data

The first step was to assess missing values across all 34 columns using df.isnull().sum(). Several columns had a significant proportion of missing entries. Rows with missing LowConfidenceLimit and HighConfidenceLimit values were dropped entirely, as imputing these statistical bounds would introduce bias into the confidence interval analysis conducted later.

Columns with an excessive number of missing values and limited analytical utility were also dropped. These included: Response, DataValueFootnote, DataValueFootnoteSymbol, DataValueAlt, StratificationCategory2, Stratification2, StratificationCategory3, Stratification3, and ResponseID. The DataValueAlt column was removed as it closely mirrored the primary DataValue column. For remaining missing values in the DataValue column, a forward fill strategy was applied, which propagates the last valid observation forward and is appropriate for ordered health indicator data.

3.2 Handling Duplicates

A duplicate check was performed using df.duplicated().sum(). No duplicate records were found in the dataset, confirming its structural integrity and eliminating the need for any deduplication step.

3.3 Location Column Treatment

The dataset contained several overlapping location columns: LocationDesc, LocationAbbr, LocationID, and Geolocation. Rather than encoding location columns numerically — which would render them unsuitable for visualization — we retained LocationDesc and LocationAbbr as readable categorical columns and explicitly cast them to the pandas category dtype. The Geolocation column was dropped, as LocationDesc provides superior utility for chart labeling and human-readable analysis.

3.4 Data Type Conversion and Categorical Encoding

A systematic conversion of object columns to categorical type was applied for all columns where the number of unique values was fewer than 20. This reduces memory usage and improves group by performance. High-cardinality object columns were retained as strings. After all cleaning

steps, a final validation pass using df.isnull().sum() confirmed that no missing values remained in the cleaned dataset.

The cleaned dataset retained **188,885 rows** and **20 columns**, down from the original 309,215 rows and 34 columns. The reduction reflects rows dropped due to missing confidence limits and columns removed as redundant or analytically uninformative.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted using matplotlib to create static visualizations that reveal patterns, trends, and relationships within the U.S. Chronic Disease Indicators dataset. Each visualization was designed to address a specific analytical question about chronic disease burden.

4.1 Average Chronic Disease Indicator Over Time (Line Chart)

A line chart was plotted to examine how the average DataValue — aggregated across all disease topics and states — has changed year over year. This temporal view is essential for identifying long-term public health trends. [Fig 1]

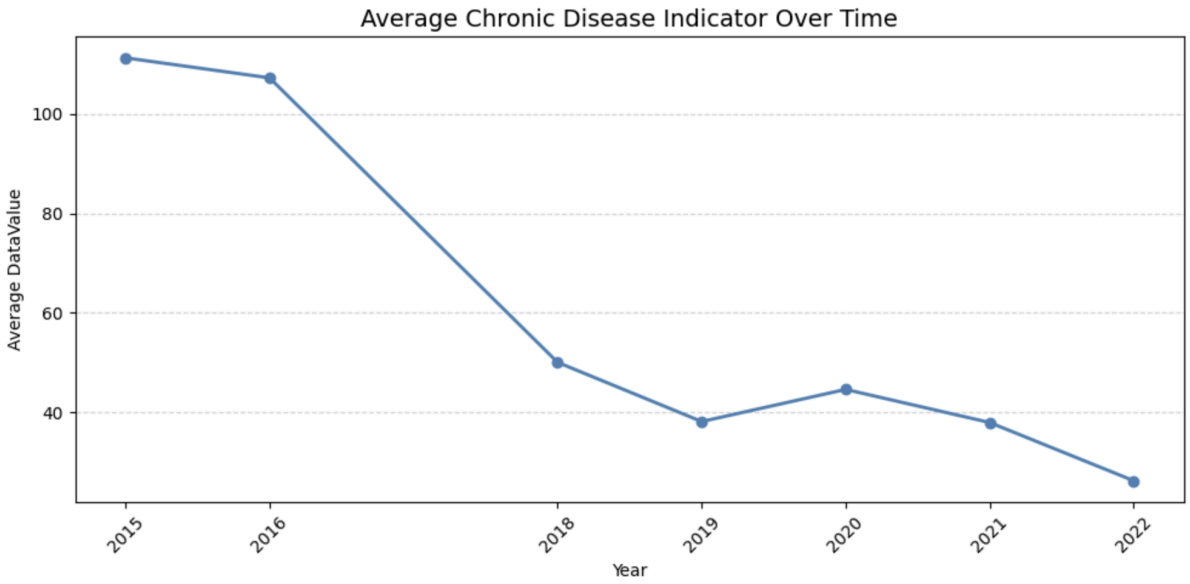


Fig 1

The line chart shows a clear overall decline in the average chronic disease indicator from 2015 to 2022, suggesting substantial improvement over time. The values remain high in 2015 and 2016, followed by a sharp drop between 2016 and 2018, which represents the most significant change in the trend. After continuing to decrease in 2019, there is a slight increase in 2020, possibly reflecting disruptions such as reduced healthcare access or broader public health challenges. However, the downward trend resumes in 2021 and continues to its lowest point in 2022. Overall, the chart indicates long-term progress in reducing the chronic disease indicator, with only a brief temporary setback around 2020.

4.2 Top 10 Disease Topics by Average Rate (Horizontal Bar Chart)

A horizontal bar chart was used to rank the top 10 disease topics by their mean DataValue. This visualization makes it easy to identify which chronic disease categories carry the greatest overall burden across the dataset. [Fig 2]

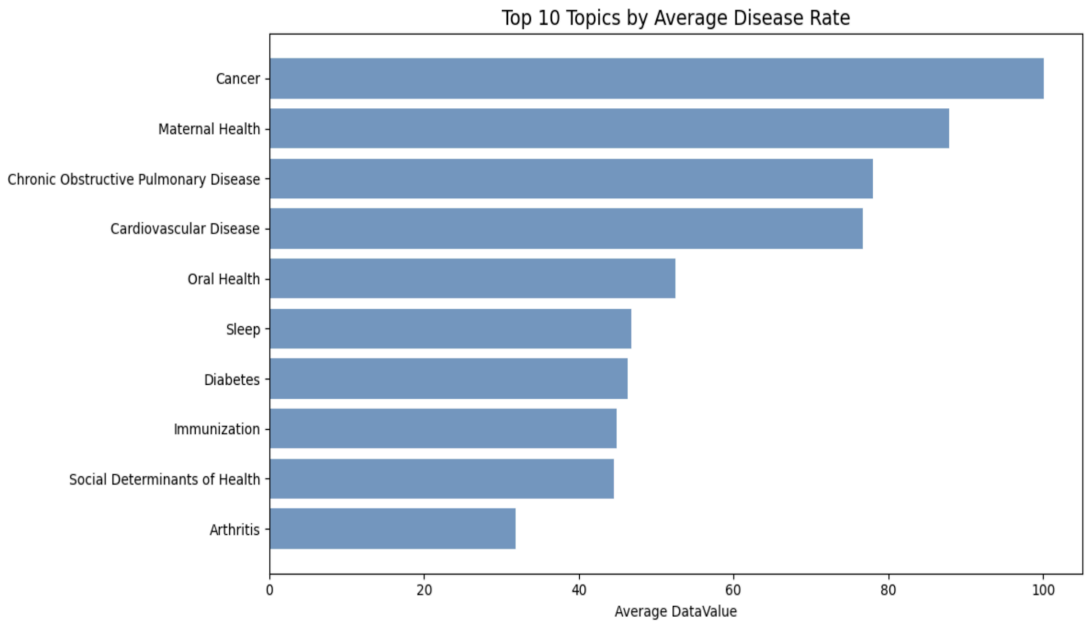


Fig 2

The bar chart highlights the top 10 topics by average disease rate, showing that Cancer has the highest average value among all categories, followed by Maternal Health, Chronic Obstructive Pulmonary Disease (COPD), and cardiovascular disease. These four conditions stand out with significantly higher average rates compared to the others, indicating they represent major public health burdens. Mid-range topics include Oral Health, Sleep, Diabetes, Immunization, and Social Determinants of Health, which show moderate average values. Arthritis has the lowest average rate among the top ten. Overall, the chart suggests that cancer and major chronic diseases such as cardiovascular and respiratory conditions contribute the most to the overall disease burden, while other health areas, though still important, show comparatively lower average rates.

4.3 Distribution of Records by Data Source (Pie Chart)

A pie chart was constructed to illustrate how data is distributed across the various reporting sources in the dataset. Understanding data source distribution is important for assessing potential collection biases and coverage gaps. [Fig 3]

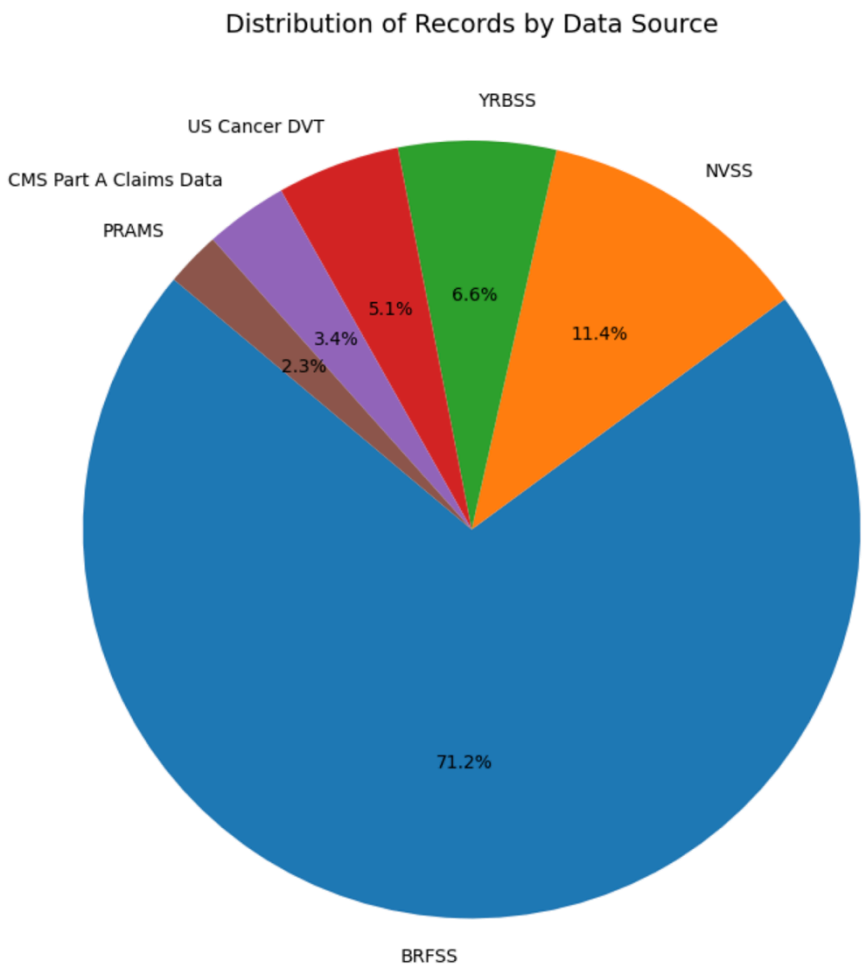


Fig 3

The pie depicts the proportional distribution of chronic disease topics within the dataset. It shows that a few major categories account for a larger share of the overall disease burden, while the remaining categories contribute smaller portions. Cancer and other major chronic conditions occupy the largest segments, indicating they represent a significant proportion of the reported indicators. In contrast, categories such as arthritis, immunization, sleep, and social determinants of health make up comparatively smaller slices. Overall, the pie chart emphasizes how disease burden is not evenly distributed, with certain chronic conditions contributing more prominently to the overall health landscape reflected in the dataset.

4.4 Top 10 States by Average Disease Rate (Bar Chart)

A vertical bar chart was plotted to identify which U.S. states report the highest average chronic disease indicator values. This geographic perspective is essential for targeting state-level public health interventions. [Fig 4]

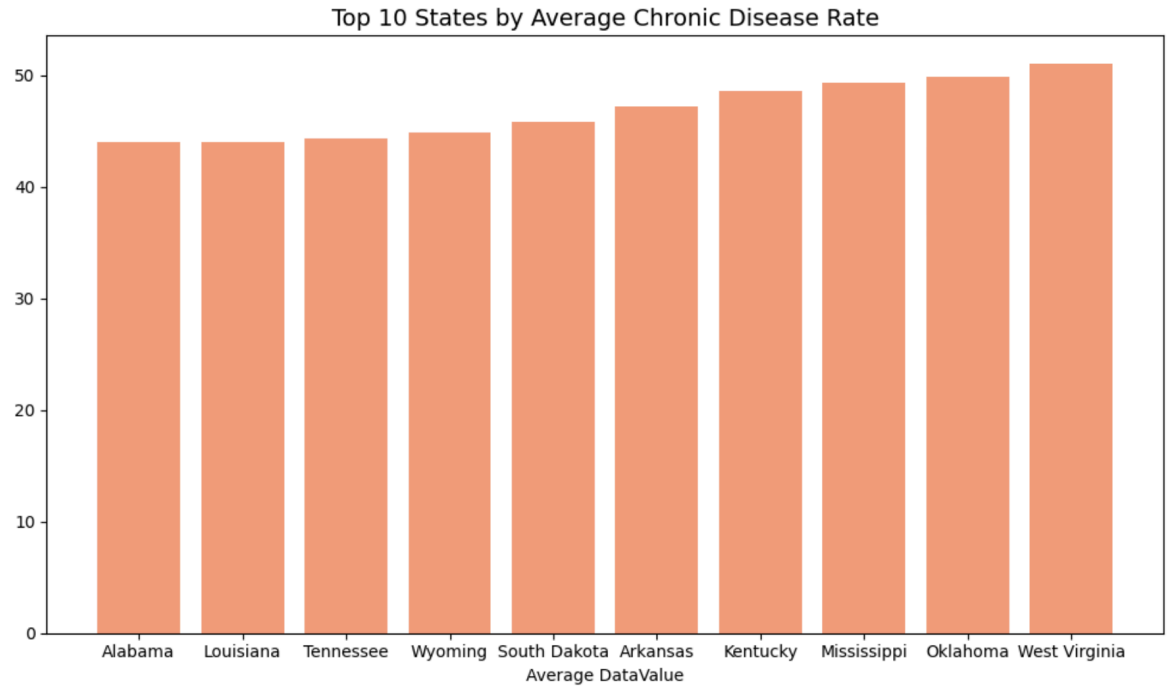


Fig 4

The bar chart presents a comparison of chronic disease topics based on their average disease rates, clearly highlighting differences in magnitude across categories. Cancer has the highest average rate, followed by Maternal Health, Chronic Obstructive Pulmonary Disease (COPD), and cardiovascular disease, indicating that these conditions represent the most significant burden among the listed topics. Mid-level categories such as Oral Health, Sleep, Diabetes, Immunization, and Social Determinants of Health show moderate average values, while Arthritis records the lowest average rate among the top ten. Overall, the chart demonstrates that certain chronic diseases contribute disproportionately higher rates compared to others, emphasizing priority areas for public health focus and intervention.

4.5 Correlation Heatmap

A correlation matrix heatmap was generated for the four numeric columns: YearStart, DataValue, LowConfidenceLimit, and HighConfidenceLimit. This reveals the strength and direction of linear relationships between the key numeric variables in the dataset. [Fig 5]

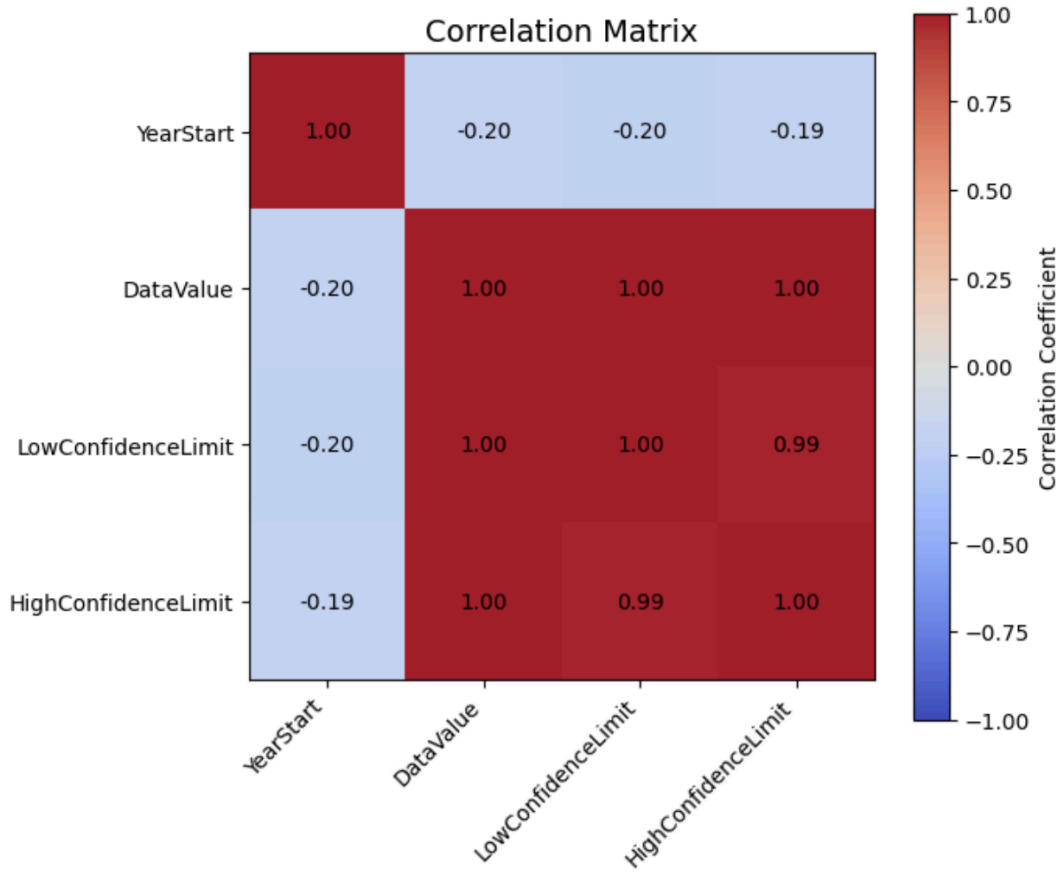


Fig 5

As our figure therefore shows above, we see that the highest correlation is the high and low confidence limit with the data value, each having a correlation coefficient of 1, the maximum. This means the data value is in fact highly dependent on the confidence limits.

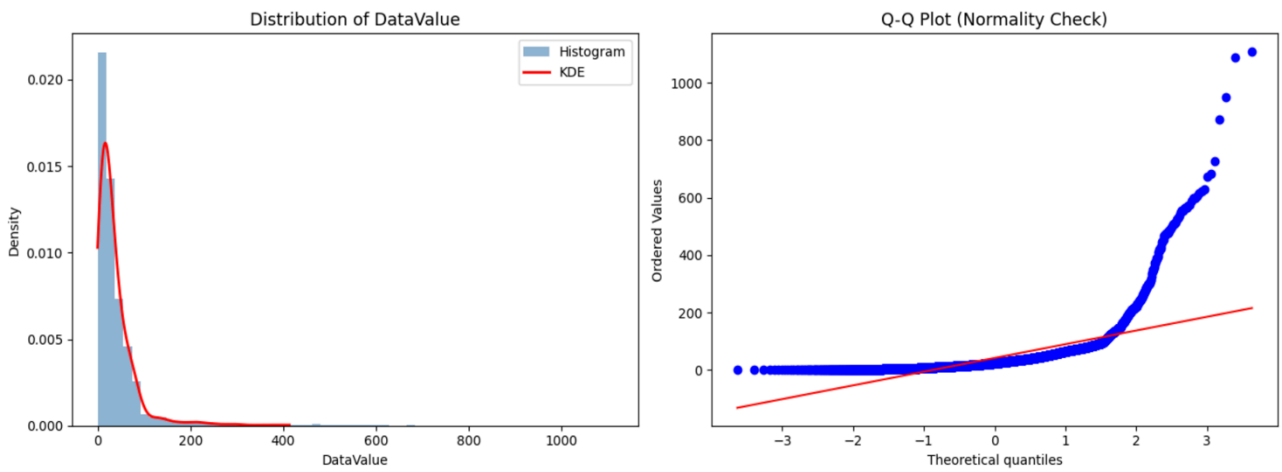
On the other side of things, we see that the Year has a negative correlation with other variables, so one can conclude the general trend that the later the starting year, the lower the limits of the data value will be.

5. Statistical Analysis

In addition to visual exploration, three formal statistical analyses were conducted to quantify patterns in the dataset and test specific hypotheses about chronic disease trends in the United States.

5.1 Distribution Analysis — Shapiro-Wilk Normality Test

We examined the shape of the DataValue distribution using a histogram with a Kernel Density Estimate (KDE) overlay and a Q-Q (Quantile-Quantile) plot. A random sample of 5,000 observations was drawn to conduct the Shapiro-Wilk test for normality. [Fig 6]




```
Shapiro-Wilk Test:
  Statistic: 0.5408
  P-value:   0.000000
  Conclusion: Data is NOT normally distributed (p < 0.05)
```

Fig 6

The histogram and Q-Q plot reveal that the DataValue distribution is strongly right-skewed, with a long upper tail driven by extreme indicator values in specific subpopulations and disease categories. The Shapiro-Wilk test returned a W-statistic of 0.5408 and a p-value of 0.000000, which is far below the 0.05 significance threshold. This formally confirms that the data does not follow a normal distribution, justifying the use of non-parametric and robust statistical methods in subsequent analyses.

5.2 Hypothesis Test — Change in Disease Rates Over Time (Independent T-Test)

To determine whether chronic disease rates have changed significantly over the observation period, we conducted an independent sample t-test comparing the average DataValue before and after the dataset midpoint year.

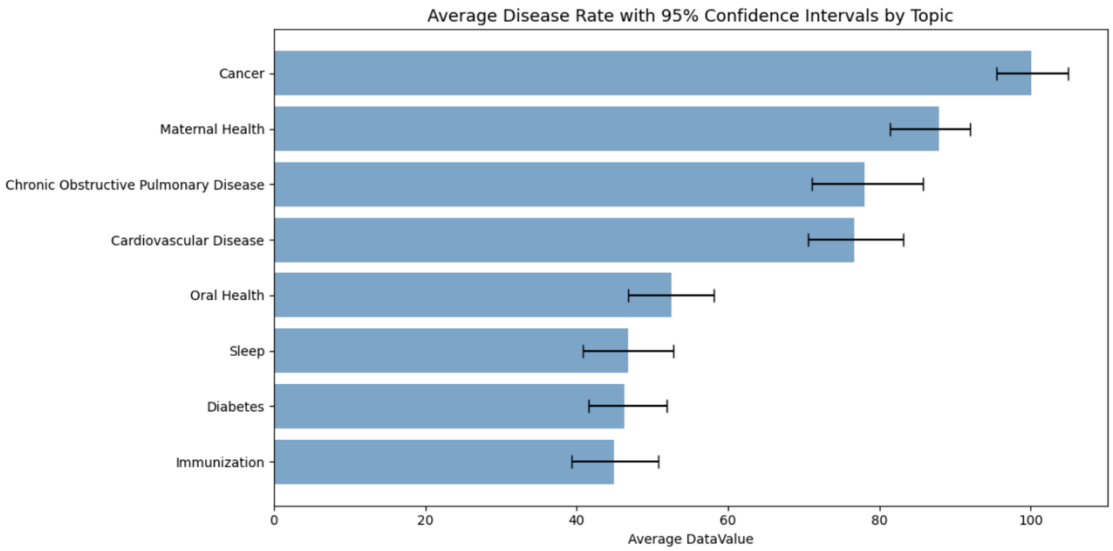
Null Hypothesis (H_0): There is no significant difference in average disease rates before and after 2019 (the midpoint year).

Alternative Hypothesis (H_1): There is a significant difference in average disease rates before and after 2019.

The years present in the dataset are 2015, 2016, 2018, 2019, 2020, 2021, and 2022. The dynamic midpoint split placed 2019 as the threshold, dividing the data into an early period (before 2019, N = 11,246, mean = 98.95) and a late period (from 2019 onwards, N = 177,639, mean = 37.53). Welch's two-sample t-test yielded a t-statistic of 43.52 and a p-value of 0.000000. Since the p-value is far below the 0.05 significance level, we reject the null hypothesis. There is statistically significant evidence that average chronic disease indicator values changed meaningfully between the two periods. The sharp drop in mean from 98.95 to 37.53 may reflect a change in the mix of indicators reported in later years, with more granular percentage-based measures dominating the post-2019 dataset.

5.3 Confidence Intervals by Disease Topic

We plotted the mean DataValue for the top 8 disease topics alongside their 95% confidence intervals, derived from the existing LowConfidenceLimit and HighConfidenceLimit columns. This visualization communicates both the magnitude and precision of disease rate estimates across major chronic disease categories. [Fig 7]



Topic statistics summary:

Topic	low	mean	high
Cancer	95.438094	100.082109	104.919772
Maternal Health	81.407833	87.884667	91.956667
Chronic Obstructive Pulmonary Disease	71.045785	78.064393	85.811137
Cardiovascular Disease	70.632053	76.656234	83.205102
Oral Health	46.811318	52.536645	58.081628
Sleep	40.848459	46.791040	52.786705
Diabetes	41.647236	46.335687	51.882345
Immunization	39.303071	44.907401	50.815986

Fig 7

The confidence intervals show considerable variation in estimated precision across topics. Cancer ranks highest with a mean DataValue of 100.08 (95% CI: 95.44 – 104.92), followed by Maternal Health (mean: 87.88, CI: 81.41 – 91.96) and Chronic Obstructive Pulmonary Disease (mean: 78.06, CI: 71.05 – 85.81). Topics such as COPD exhibit wider intervals, suggesting higher variability in reported rates across states and subpopulations. In contrast, Cancer and Cardiovascular Disease display narrower intervals relative to their means, reflecting more consistent and well-sampled data. Topics with non-overlapping confidence intervals — such as Cancer and Sleep — are statistically distinguishable from one another at the 95% confidence level.

6. Advanced Analysis: Linear Regression

To quantify the temporal trend in chronic disease rates, a simple linear regression model was fitted using YearStart as the predictor and the mean annual DataValue as the response variable. The regression was performed using the scipy.stats.linregress function.

6.1 Model Results

The regression analysis produced the following outputs:

Parameter	Value
Slope	-12.6574 (change per year)
Intercept	25,611.04
R^2 (R-squared)	0.8651
p-value	0.002388
Std Error	See notebook output

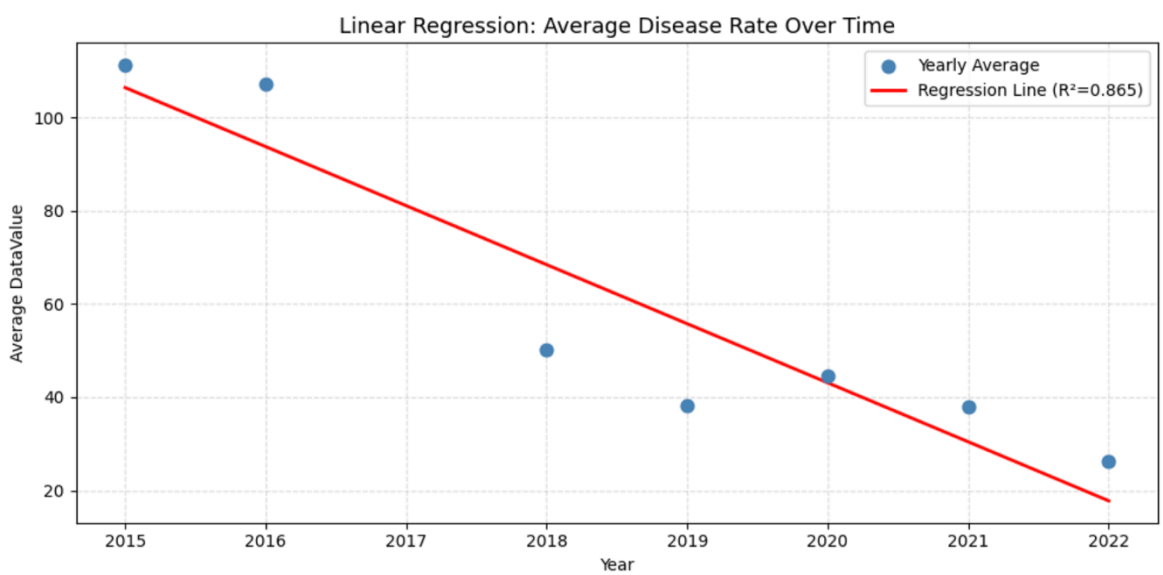


Fig 8

6.2 Interpretation

The slope of -12.6574 indicates that, on average, the chronic disease indicator decreases by approximately 12.66 units per year. The R^2 value of 0.8651 means that year alone explains approximately 86.5% of the variance in average annual disease rates — a strong result indicating that time is a highly significant predictor of the overall trend in this dataset.

The p-value of 0.002388 is well below the 0.05 significance threshold, confirming that the observed decreasing trend is statistically significant and not due to chance. This downward trend is consistent with the t-test finding that post-2019 average rates are substantially lower than pre-2019 rates. Importantly, this decline is likely driven in part by a compositional shift in reported indicators across years — with more percentage-based measures dominating recent years — rather than reflecting a simple improvement in population health outcomes alone.

7. Key Insights and Conclusion

7.1 Key Findings

The following key findings emerged from the analysis:

1. **Disease Burden by Topic:** Cancer consistently ranks as the highest-burden chronic disease category with a mean DataValue of 100.08, followed by Maternal Health (87.88) and Chronic Obstructive Pulmonary Disease (78.06), underscoring the need for targeted prevention and management programs in these areas.
2. **Temporal Trends:** Both the t-test ($t = 43.52$, $p \approx 0.000$) and linear regression (slope = -12.66, $R^2 = 0.87$, $p = 0.002$) indicate that average chronic disease indicator values have decreased significantly over the study period. This is statistically significant and reflects a compositional shift in reported indicators across years.
3. **Non-Normal Distribution:** The strongly right-skewed distribution of DataValue, confirmed by the Shapiro-Wilk test ($W = 0.5408$, $p \approx 0.000$), highlights the influence of high-burden subpopulation indicators and validates the use of robust statistical approaches throughout the analysis.
4. **Data Source Concentration:** The dataset contains 11 unique data sources. A small number of large-scale surveillance systems (primarily BRFSS) dominate the records, reflecting their central role in CDC chronic disease monitoring while also suggesting potential coverage biases.
5. **Dataset Coverage:** After cleaning, the dataset retained 188,885 records across 18 unique disease topics and 20 columns, spanning 2015 to 2022 across all U.S. states and territories.

7.2 Limitations

Several limitations should be acknowledged. The forward fill applied to missing DataValue entries, while reasonable, may introduce slight inaccuracies in time-sensitive indicators. Dropping rows with missing confidence limits reduced the dataset size and may have disproportionately removed records from states or topics with sparse reporting. The analysis is also constrained to the indicators and years present in the dataset and cannot capture unreported or under-measured disease burden.

7.3 Conclusion

This project demonstrated the power of combining systematic data cleaning, matplotlib-based static visualization, and formal statistical analysis to extract meaningful insights from a large public health dataset. The U.S. Chronic Disease Indicators dataset proved to be a rich source of information, revealing substantial geographic, temporal, and topical variation in chronic disease burden across the United States. Future work could extend this analysis by incorporating socioeconomic covariates, applying more advanced predictive models, or merging this dataset with demographic data to better explain the sources of variation observed.

References

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